CSC 350 Assignment 4
Spring 2025

Due: Tuesday, March 4 by 11:59 pm

General Instructions

- Scan or compile snapshots of your submission into a PDF and upload it to Canvas.
- Assignments are **due before midnight** on the scheduled due date; generally, late assignments will not be accepted.
- Unless specified, **justify your answers always**. If justifications are missing or do not show reasonably how your answers are derived, 0 points will be assigned regardless of the correctness of the answers.
- Write **neatly** in the event you write your answers on paper. **Especially, for table completion questions, if your answers are not given neatly in well-drawn tables, the answers won't be graded**.
- **This assignment must be done by yourself without any unauthorized assistance**. Write your full name and numeric university ID on the front page of your submission, along with the following pledge statement.

I pledge my honor that all parts of this assignment were done by me individually and without any unauthorized assistance or collaboration with others.

Q1. Consider the equivalence rule 7.2.4b of the textbook.

Prove $\forall x\,(p(x) \wedge q(x)) \equiv \forall x\,p(x) \wedge \forall x\,q(x)$ from 7.2.4b using other equivalences.

Hint: Start with the left-hand-side of 7.2.4b, and use other equivalences to transform the lhs to equivalent formulas until you obtain the right-hand side of 7.2.4b (Or, you can start with the right-hand-side and obtain the left-hand-side). Write down your work step by step very clearly, stating which equivalence you are using for each step.

Q2. Consider the equivalence rule 7.2.9b of the textbook.

Prove $\exists x\,(C \rightarrow A(x)) \equiv C \rightarrow \exists x\,A(x)$ from 7.2.9b using other equivalences. That is, start with the left-hand-side of 7.2.9b, and use other equivalences to transform the left-hand-side to equivalent formulas until you obtain the right-hand-side of 7.2.9b (or vice-versa). Note that x does not occur free in C in this equivalence. Write down your work step by step very clearly, stating which equivalence you are using for each step.

Q3. Repeat Q2. for equivalence 7.2.9d.

Prove $\exists x\,(A(x) \to C) \equiv \forall x\,A(x) \to C$ from 7.2.9d using other equivalences.

Q4. Prenex Normal Form

Consider the following formula, in which $p(x, y)$ means x is a parent of y and $g(x, y)$ means x is a grandparent of y.

$$\forall x \forall y\,(\exists z\,(p(x, z) \wedge p(z, y)) \to g(x, y))$$

(a) Explain, in English, what the formula means.

(b) Use equivalences to construct a prenex normal form for the above wff. Show your work. Write down the number and name of the equivalence used at each step.

Q5. Prenex DNF

Use equivalences to construct a prenex disjunctive normal form for the following wff. Show your work. Write down the number and name of the equivalence used at each step.

$$\forall x \exists y\,p(x, y) \to \exists y \forall x\,p(x, y)$$

Q6. Prenex CNF

Use equivalences to construct a prenex conjunctive normal form for the following wff. Show your work. Write down the number and name of the equivalence used at each step.

$$\forall x \forall y \forall z\,(p(x, y) \wedge p(y, z) \to p(x, z)) \wedge \forall x \neg p(x, x) \to \forall x \forall y\,(p(x, y) \to \neg p(y, x))$$